

Shwetha Prakash

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Research Interests

Cosmology; CMB and mm-wave instrumentation; KID detector simulations; radio interferometry; astronomical data analysis; numerical astrophysics.

Education

Cornell University

Ph.D. in Physics (Year 2)

Ithaca, NY

2024–Present

Ashoka University

Postgraduate Diploma (Advanced Major: Physics, Minor: Astronomy), GPA: 3.73/4.0

Sonapat, India

2024

B.Sc. (Honours) Physics, GPA: 3.53/4.0

2023

Current Research

Advisors: Dr. Abigail Crites, Dr. Michael Niemack

TIME (Tomographic Ionized Carbon Mapping Experiment)

- Optimizing the pipeline that performs calibration, reduction, and analysis of data collected by TIME.
- Instrumentation of the cryo system, TES detectors and on site commissioning runs at Kitt Peak.

CCAT - PrimeCAM

- Developing software tools for data acquisition from mKID detectors on PrimeCAM.
- Exploring characterisation and data analysis of large mKID arrays.

Technical Skills

Programming: Python (NumPy, SciPy, Matplotlib, AstroPy, GWpy), LaTeX

Data Analysis: sotodlib (Simons Observatory), CASA, IRAF, HEASoft, DS9, Siril, friendlyVRI

Scientific Computing: PLUTO MHD code, General Particle Tracer (GPT), Jupyter, Docker (learning)

Version Control: Git, GitHub

Instrumentation: GSO StarTracker, Celestron NexStar 7SE telescope, STC-7 CMOS camera

Operating Systems: Linux, macOS, Windows

Teaching Experience

Cornell University

- PHYS 2207: Fundamentals of Physics I Teaching Assistant

Fall 2024, Fall 2025

- PHYS 2208: Fundamentals of Physics II Teaching Assistant

Spring 2024

Ashoka University

- Observing the Cosmos (Astronomy Lab) Teaching Assistant

Spring 2023

- Classical Mechanics and Electromagnetism Lab Teaching Assistant

Fall 2023

Past Research Experience

Senior Thesis Ashoka University

Aug 2023–May 2024

Mentors: Dr. Bikram Phookun, Dr. Nissim Kanekar

- Performed radio interferometric analysis of Damped LymanAlpha systems using VLA HI emission data.
- Developed CASA-based workflows for calibration and imaging.

CLASSE REU Cornell University

Jun 2023–Aug 2023

Mentor: Dr. Jared Maxson

- Simulated and built a vacuum-compatible solenoid for electron-beam focusing in UED experiments.
- Conducted COMSOL and GPT simulations for beam transport optimization.
- Featured in Cornell media:
[Project Summary](#),
[REU Interview](#).

Advanced Independent Lab (Soft Matter Physics)

Jan 2023–May 2023

Mentor: Dr. Pramoda Kumar

- Fabricated a liquid crystal cell and studied its birefringence experimentally.

Selected Projects

Monte Carlo Simulations of Lennard–Jones Systems

Aug 2022–Dec 2022

- Implemented Monte Carlo algorithms in Python to explore thermodynamics of lattice systems.

Earths Mass and J_2 Perturbation via Orbital Simulation

Jun 2022–Aug 2022

- Modeled satellite precession using RK4 methods and AstroSAT data to estimate Earths J_2 .

Solar Temperature from Spectral Analysis

Nov 2021–May 2022

- Designed a spectrometer and analyzed solar spectra using Wiens displacement law.

Modeling Chladni Figures via Finite Difference Methods

Jan 2022–May 2022

- Simulated 2D wave equations in Python to reproduce vibrational modes of plates.

Leadership and Service

Vice President Physics Graduate Society, Cornell University

2025–Present

- Coordinating academic, social, and mentorship events for the Cornell physics graduate community.

Founding President Astronomy Society, Ashoka University

2023–Present

- Organized Indias first Undergraduate Radio Astronomy Conference.
- Led workshops on optical data reduction and telescope operation.

Core Team Ashoka University Women in STEM (AUWS)

2023–Present

- Advocated for integrating women's STEM engagement into the LGP curriculum.

Head Astronomy Club, Ashoka University

2022–2023

- Organized a three-day astronomy fest and hands-on instrumentation sessions.

Founder and Head of Design Samatvam India

2020–Present

- Co-founded an NGO promoting gender and social equality through science outreach.